Self-consistent electromagnetic feedback in Majorana nanowires: a validated single-channel null result, and its breakdown beyond one subband

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Standard treatments of the Majorana nanowire fix the topological phase boundary and compute the induced supercurrent afterward; the current is never fed back into the Hamiltonian that produced it. We close this loop with a validated framework – a particle-hole-symmetric Bogoliubov-de Gennes eigensolver, a real-space Pfaffian $\mathbb{Z}_2$ invariant, and a Biot-Savart ring self-field – so that the wire's self-generated flux dresses the Peierls phase entering its own Hamiltonian. In the single-channel limit this feedback produces no genuine bulk shift of the topological boundary at any dynamically accessible coupling strength; the null result is robust under finite-size scaling, ring-inductance variation, operating temperature up to and beyond 20 mK, and on-site disorder at the lower end of a literature-grounded range. Two results recur throughout: a conventional spectral-gap threshold detector overstates the shift by an order of magnitude relative to the threshold-free invariant, and the self-consistency loop has a dynamical bifurcation above which a fixed point exists but is not Picard-attracting, so existence and dynamical stability must be verified independently. Extending the calculation to a second occupied transverse subband overturns the null: a pre-registered loop-gain diagnostic crosses the Picard stability threshold at the production electromagnetic coupling once $M \geq 2$, and a full self-consistent boundary calculation at $M = 2$ shows that the *existence* of a static topological boundary becomes fragile to the lattice discretization used to compute it. Extending the same diagnostic to $M = 5, 6$ shows this trend is itself non-monotonic, not a simple worsening with occupancy: the loop-gain statistic collapses at $M = 6$, but the underlying detachment rate – whether any static solution exists at all – stays high and non-monotonic throughout, so existence, not the loop-gain magnitude, remains the operative diagnostic. Because realistic InAs/Al devices are estimated to occupy far more than two subbands, this existence-fragility result – not the single-channel null – should govern how electromagnetic feedback is treated in real devices. Code, validation scripts, and raw data are publicly available.

I. INTRODUCTION

A one-dimensional topological superconductor supporting unpaired Majorana zero modes (MZMs) at its ends was first shown by Kitaev to arise from a spinless p-wave pairing model whose bulk topology is captured by a $\mathbb{Z}_2$ invariant [1]. Oreg, Refael, and von Oppen, and independently Lutchyn, Sau, and Das Sarma, showed that the same physics is realized in a much more accessible platform: a semiconducting nanowire with strong Rashba spin-orbit coupling, placed in an external Zeeman field and proximitized by an s-wave superconductor [2, 3]. This proposal launched over a decade of experimental work on InAs/Al and InSb/Nb hybrid nanowires, whose appeal for fault-tolerant quantum computation rests on the non-Abelian exchange statistics of MZMs and the resulting topological protection of information encoded in their collective fermion parity [4].

The experimental record is real but contested. Zero-bias conductance peaks – the original proposed signature – have repeatedly been shown to also arise from disorder-induced trivial Andreev bound states and other non-topological mechanisms indistinguishable from a genuine Majorana signature in tunneling conductance alone [5, 6], and claims in this field have a documented history of being revised or withdrawn under closer scrutiny. Progress has continued: a 2025 measurement from Microsoft Quantum reported a single-shot interferometric readout of fermion parity in an InAs-Al hybrid device, with a reported assignment error of 1%, framed explicitly as a step toward – not a demonstration of – topological qubits [7]. That same measurement has since been the subject of a direct methodological challenge: a comment by Legg [8] argues that the topological gap protocol used to certify the device's gapped, disorder-free operating regime is unreliable and that publicly available data in the same regions used for the parity readout is, on direct inspection, disordered with no clear gap. We report this dispute as part of the field's current state, not as a resolved matter either way. Distinguishing genuine topological physics from look-alike trivial physics in a disordered, finite device remains the central open problem in this field – indeed, recent work argues that in short, disordered wires the topological classification itself may not be uniquely defined and requires context-dependent analysis [9] – and it is the reason a validated, threshold-free topological diagnostic – not a spectral proxy – is a prerequisite for any claim made in this paper.

A second, largely separate line of question concerns the wire's electromagnetic environment. Realistic Majorana device architectures embed the nanowire in a circuit – gate electrodes, coplanar-waveguide resonators for dispersive readout, and the loop inductance of the proximitizing superconductor itself. The current flowing in the wire, set self-consistently by the wire's own quantum state, generates a magnetic self-field; if that field is large

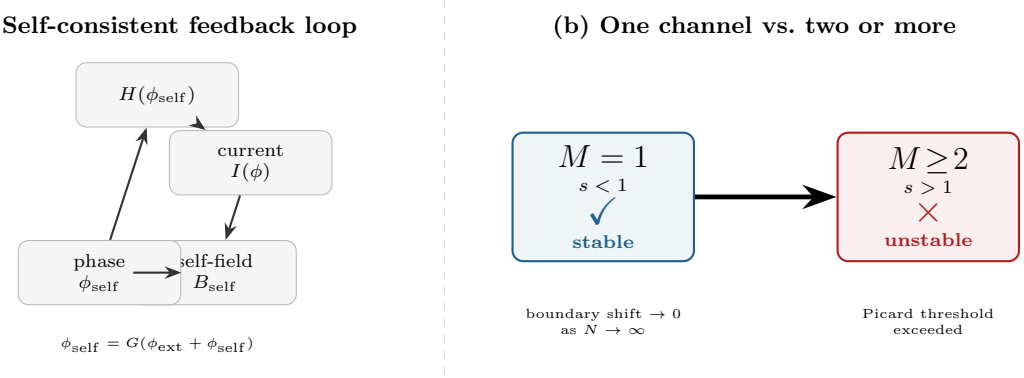


FIG. 1. **Graphical summary.** (a) The loop closed in this paper: a bond-phase texture fixes H, whose ground-state current sources a Biot-Savart self-field, whose Peierls phase feeds back into H. (b) The central result: at $M = 1$ the loop-gain slope stays sub-critical ($s < 1$, validated null); once a second transverse subband is occupied ($M \geq 2$), s crosses the Picard stability threshold at production coupling and the static topological boundary itself becomes fragile.

enough, and slow enough on the relevant timescale, it reenters the Hamiltonian as a phase twist and could in principle shift the very topological phase boundary the circuit is built to read out. To our knowledge, no prior published work has closed this loop for the electromagnetic (inductive, current-generated flux) channel specifically together with a genuine, threshold-free topological invariant. The closest related work addresses different feedback channels or a different role for inductance entirely: Vuik *et al.* treat a self-consistent *electrostatic* (Coulomb, Schrödinger-Poisson) response of the chemical potential to an applied field and show it reshapes the topological phase boundary [10]; Lobo, Prada, and San-Jose treat a self-consistent Hartree-Fock-Bogoliubov *pairing potential* in intrinsic (non-hybrid) wires [11]; Shustin, Aksenov, and Burmistrov compute many-body *charge-fluctuation* corrections to the topological winding number via a Green's-function self-energy, distinct from any classical electromagnetic self-field [12]; and Roy, Sau, and Tewari propose quantum inductance as a phase-sensitive *diagnostic probe* of a fixed, non-self-consistent Hamiltonian [13], not as a mechanism that modifies the boundary itself. Our work is the electromagnetic-feedback analogue of Vuik *et al.*'s electrostatic-feedback question, and we adopt the same caution in framing it: the answer we find is a null result in the regime we can fully validate, and a non-null, unresolved result beyond it, not a clean confirmation in either direction.

We build and validate a computational framework that closes this loop for a Lutchyn-Oreg nanowire on a ring geometry (§II) and report a findings arc organized around two axes: robustness of the single-channel result, and its extension to a second occupied transverse subband (§III). The single-channel result is a validated null: no bulk topological-boundary shift survives finite-size scaling, variation of the wire's own inductance, realistic operating temperature, or (at the lower end of the literature-realistic range) explicit on-site disorder. Two methodological lessons recur throughout this arc and are central to it: first, that a conventional spectral-gap threshold detector is not a reliable proxy for the bulk topological invariant, even at fixed system size, and produces an order-of-magnitude-overstated boundary shift; second, that a self-consistency loop of this kind possesses a genuine dynamical bifurcation, above which mathematical fixed points continue to exist but are no longer attracting under the physical relaxation dynamics – so existence and dynamical stability of a solution must always be checked as two separate questions. Extending the calculation to two occupied transverse subbands overturns this null result, and does so in a way that is qualitatively more severe than a magnitude change: at the production electromagnetic coupling, the loop gain crosses the stability threshold once two or more subbands are occupied, and a full self-consistent boundary calculation at a representative two-subband operating point shows that the very existence of a static topological boundary becomes fragile to the lattice discretization used to compute it. Given that realistic InAs/Al devices are estimated to occupy far more than two subbands [14, 15], we present this existence-fragility result, not the single-channel null, as the paper's operative conclusion for real devices (§IV).

Every result in this paper – including intermediate results that were later superseded by more rigorous checks – is preserved and traceable in a public code repository, together with every validation script and raw output file; see the data availability statement at the end of this paper.

II. MODEL AND COMPUTATIONAL FRAMEWORK

A. Lutchyn-Oreg Hamiltonian and material parameters

We use the standard real-space, tight-binding Lutchyn-Oreg BdG Hamiltonian for a single conducting channel: a

chain of N sites with lattice constant a, nearest-neighbor hopping t, Rashba spin-orbit coupling α, chemical potential μ, in-plane Zeeman field B, and s-wave proximity pairing Δ, written in the Nambu basis (electron/hole $\otimes$ spin $\otimes$ site, dimension $4N$) so that the hole block is the exact conjugate of the electron block and particle-hole (PH) symmetry $\Sigma_x H^* \Sigma_x = -H$ holds identically for any parameter choice, including a spatially non-uniform phase texture on the bonds. Energies are quoted in units of Δ unless stated otherwise. Material parameters are drawn from InAs bulk properties ($m^* = 0.023\, m_e$, g-factor 10.5, $\alpha_{\mathrm{SO}} = 20$ meV nm) [2, 3] and from the induced gap directly measured in epitaxial-Al/InAs full-shell nanowires, $\Delta = 0.19$ meV [15], giving $t \approx 66.3\,\Delta$ and $\alpha \approx 8.0\,\Delta$ at $a = 10$ nm. Chang *et al.* additionally report a base operating temperature of 20 mK and an axial critical field of ≈ 75 mT for a comparable thin epitaxial-Al shell [15]; the latter is well below the Zeeman field needed to reach this model's topological transition at these parameters, so the model's own pairing potential Δ is held fixed (no field-dependent pair-breaking $\Delta(B)$), an idealization we carry explicitly through every figure and result that sweeps B into that regime.

B. Topological invariant: a threshold-free real-space Pfaffian

Rather than detecting the topological transition as the field at which the lowest BdG excitation crosses a fixed absolute energy threshold, we use the genuinely discrete $\mathbb{Z}_2$ invariant: the sign of the Pfaffian of the Hamiltonian rotated into its Majorana (real antisymmetric) basis [1]. For a periodic ring we use the periodic-boundary-condition (PBC) Pfaffian alone rather than the full PBC×antiperiodic-boundary-condition (APBC) product used in the general real-space construction [16]: in this regime the $k = \pi$ gap never closes near the transition and contributes only a constant, non-flipping sign, while $k = 0$ is an allowed ring momentum at every system size N, so the boundary located this way carries *zero* measured finite-size shift, in contrast to the conventional spectral threshold detector. We validate this construction against the known analytic boundary at multiple chemical potentials, ring sizes, and bisection tolerances, and against a threshold detector swept over its own threshold at fixed N (procedural detail in Appendix B; outcomes summarized in the first three rows of Table IV). Results are given in §III A.

A companion correction is recorded here as part of this framework's validation discipline: an earlier module computing this same invariant never actually called the Pfaffian routine it was supposed to, and the bug survived undetected only because of a parameter regime in which a hardcoded closed-form stand-in happened to agree with it. It was caught, fixed, and re-validated with zero disagreements against the analytic boundary across a 1600-point sweep before any result relying on it was trusted; full mechanism detail is recorded in the correction ledger (Appendix C, item 2).

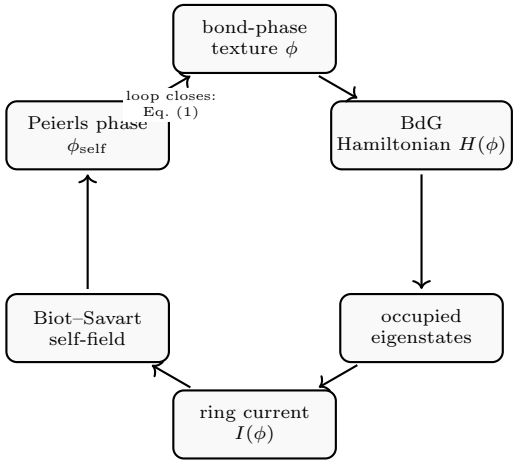


FIG. 2. **The feedback loop this paper closes: current sources the very field that re-enters its own Hamiltonian.** Conceptual diagram, not a data plot. A bond-phase texture fixes the BdG Hamiltonian, whose occupied eigenstates carry a ring current; that current sources a Biot-Savart self-field, which dresses the hopping bonds with an additional Peierls phase that feeds back into the texture the loop started from. The loop closes at the fixed-point condition of Eq. (1), $\phi_{\mathrm{self}} = G(\phi_{\mathrm{ext}} + \phi_{\mathrm{self}})$.

C. Self-consistent electromagnetic feedback

The physical motivation precedes the formalism. Once the Hamiltonian's ground state is fixed at some applied field, the wire carries a real, physical current circulating around the ring, and that current is a genuine magnetic-field source. If the induced self-field is large enough, and slow enough on the relevant electronic timescale to be treated quasi-statically – a separation of scales borne out experimentally, since measured quasiparticle-poisoning (parity) lifetimes in comparable proximitized hybrids reach the millisecond range [17], many orders of magnitude slower than the induced gap's own $\hbar/\Delta \sim 3.5$ ps timescale at $\Delta = 0.19$ meV – it re-enters the same additional phase on every hopping bond. This closes a feedback loop that most prior treatments of this problem leave open, by computing the induced current only after the topological phase boundary has already been fixed and never feeding it back in.

The wire is closed into a ring of circumference $L = Na$. The ring current density $I(\phi)$, summed over occupied BdG quasiparticle states at a given uniform bond-phase texture ϕ, generates a magnetic self-field via the Biot-Savart law adapted to the closed geometry (validated against the analytic thin-wire loop self-inductance to better than 0.5%), which re-enters the Hamiltonian as an additional Peierls phase on every bond. The self-

consistency condition is

$$\phi_{\rm self} = G(\phi_{\rm ext}+\phi_{\rm self}), \qquad G(\phi) = g\,\frac{e}{\hbar}\,a\,A_{\rm self}[I(\phi)]\,, \quad (1)$$

where g is a dimensionless effective electromagnetic feedback gain representing the combined action of a resonant readout circuit (quality factor Q) and the ratio of kinetic to geometric inductance in the nanowire loop. Superconducting coplanar-waveguide resonators with internal quality factors exceeding 10^6–10^7 have been demonstrated experimentally [18], and nanowire loops with large kinetic-inductance contributions relative to their bare geometric inductance have likewise been realized [19]; combining representative values from these two independent, literature-grounded mechanisms motivates treating g as a plausibility estimate spanning several orders of magnitude around $g \sim 10^4$–10^6, not a value derived from a specific circuit design. We adopt $g_{\rm production} = 2.9 \times 10^4$ as the single coupling value used throughout the production results below (§IID), and separately characterize the loop's behavior as a function of g across this entire plausibility range.

On a uniform ring the self-consistency condition reduces to a scalar fixed-point equation in ϕ. We solve it three independent ways: (i) bisection on $G(\phi) - \phi = 0$, which requires no stability property of the map and locates a root regardless of whether ordinary iteration would converge to it; (ii) plain (Picard) fixed-point iteration, which converges only where the map's local slope $s \equiv dG/d\phi|_{\phi^*}$ satisfies $|s| < 1$; and (iii) Anderson-accelerated mixing, a quasi-Newton method that can locate a root even where the bare map is Picard-repelling, gated on both a step-size criterion and an independent fixed-point-residual check so that a stalled iterate is never mistaken for a converged one. A Newton-Krylov search over the full N-dimensional (non-uniform) fixed-point equation is used to test directly whether any symmetry-broken solution exists beyond the uniform one (seeding protocol in Appendix B). This three(-plus-one)-solver redundancy is what let us distinguish, as described in §III B, a root's mathematical existence from its dynamical attractiveness – a distinction that turns out to matter throughout this paper.

D. Multi-subband extension

To go beyond a single conducting channel we extend the basis to site(N)⊗subband(M)⊗spin(2)⊗Nambu(2), dimension $4NM$ (not $4NM^2$): each subband contributes a diagonal $4N$ block at its own shifted chemical potential $\mu-\varepsilon_n$, plus a mandatory inter-subband Rashba coupling $\propto \langle n|k_y|m\rangle$ that vanishes for even $n-m$ and is otherwise fixed by the hard-wall transverse mode overlap. Omitting this term would not be a mere approximation: it is the mechanism by which multiband occupancy hybridizes Majorana pairs from different transverse modes and closes the effective gap [20], so a model that dropped it would be qualitatively, not just quantitatively, wrong. The construction is validated by four gates before any physics result is trusted: exact reproduction of the single-channel Hamiltonian and current operator at $M = 1$; exact decoupled-limit agreement with the direct sum of M single-channel blocks when the inter-subband coupling is set to zero; particle-hole symmetry at machine precision for $M = 1$–4; and an exact match of the inter-subband coupling matrix to its closed-form selection rule. All four gates pass exactly.

The construction fixes a transverse hard-wall width $W = 50\,\text{nm}$, chosen together with the van Hove exclusion tolerance below so that the exclusion radius stays a small fraction of one subband spacing at every M tested. This gives a first-subband energy $\varepsilon_1 \approx 26.2\,\Delta$ and a full subband spacing growing from $\Delta E_{\rm sb}(1) \approx 19.6\,\Delta$ to $\Delta E_{\rm sb}(6) \approx 85.0\,\Delta$ across the range tested below – large compared to the $\sim$ 2–30 meV (order a few to several tens of Δ) inter-subband spacings Woods *et al.* report across their modeled gate-charge range for InAs/Al devices [14]. The present construction therefore resolves each newly occupied channel cleanly from its neighbors, at the cost of not matching any specific real device's cross-sectional geometry – a structural, not device-faithful, choice, consistent with every other idealization carried through this study.

Because the loop-gain slope s is expected on general grounds to be structured near any subband threshold (a one-dimensional van Hove singularity in the density of states) rather than smooth in the chemical potential, we pre-register – in a separate commit, before any value of $s(\mu; M)$ is computed – a van Hove exclusion radius derived in closed form from the bare tight-binding dispersion (and verified against a genuinely diagonalized spectrum at threshold, not merely the smooth bulk), a μ-sweep window for each subband count M that straddles its own occupation threshold, and a comparison statistic (the median of $s(\mu)$ over the surviving, non-excluded window, chosen over the maximum specifically because the maximum would be pulled toward whatever point sits nearest an excluded boundary). This locked-before-outcome protocol mirrors the discipline this project applied after an earlier, unqualified coupling-strength headline turned out to describe a dynamically unreachable root (§III B): fix the analysis before the number that could be affected by it exists.

E. Disorder and finite temperature

On-site disorder is added as $\varepsilon_i \sim \text{Uniform}[-W/2, W/2]$, i.i.d. over sites, which preserves the PH-symmetry identity exactly for any realization (verified both algebraically and numerically). Every reported disordered quantity is an ensemble average over independent realizations drawn with an explicit, reproducible multiprocessing-safe seeding scheme, with the realization count and its spread reported alongside

every mean. The realistic disorder strength is grounded directly in a re-verified primary source: Ahn *et al.* report an on-site random potential with standard deviation $\sigma_\mu = 2.5$ meV for InAs/InSb Majorana nanowires [21], which converts (with the conversion factor itself flagged as the softest number in this part of the calculation, since it depends on matching a Gaussian literature disorder model to this work's uniform-distribution convention) to $W \approx 10$–$46\,\Delta$ at our Δ.

Finite temperature enters through Fermi-Dirac occupation of the BdG quasiparticle spectrum, applied to the self-field current, the boundary detector, and a finite-frequency current-current (Kubo) correlator used for the reactive-impedance diagnostic below; every finite-temperature quantity is validated to reproduce its zero-temperature counterpart bit-for-bit (or, for the Kubo correlator, to floating-point precision) as $\theta \equiv k_B T/\Delta \to 0$.

F. Reactive impedance diagnostic

As a second, independent observable alongside the bulk gap, we define a feedback-renormalized reactance $Z_r(\omega) = i\omega L_{\text{eff}}$, $L_{\text{eff}} = L_{\text{geo}}/(1-s)$, where s is the same self-consistency loop-gain slope defined above and L_{geo} is the geometric self-inductance of the ring, independently validated against the analytic thin-wire loop formula. This linearization of the closed-loop response diverges precisely at the Picard bifurcation $s \to 1$, giving a circuit-level observable sensitive to the same instability the self-consistency solver encounters directly. We cross-check its zero-frequency limit against a finite-frequency Kubo current-current correlator computed from the full BdG spectrum (a fully independent, non-reused calculation), which must reduce to L_{eff} as $\omega \to 0$; this cross-check is a required gate, not a reported result, and failing it would indicate a bug rather than new physics.

G. Data availability

The complete codebase used to produce every result in this paper – including every validation script referenced above, all raw numerical results (as JSON/CSV data files), and every figure-generating script – is publicly available in the GitHub repository accompanying this work. Superseded intermediate results (see, e.g., §III D and §III E) are preserved in the repository rather than overwritten, with the correction and its reasoning recorded alongside.

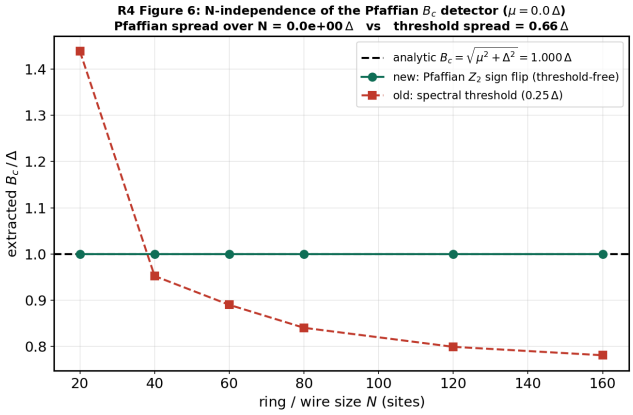


FIG. 3. **The threshold-free Pfaffian invariant is exactly N-independent; the conventional spectral-gap threshold detector is not.** N-independence of the real-space Pfaffian boundary detector versus a conventional spectral-gap threshold detector, at $\mu = 0$ (known analytic boundary $B_c = \Delta$). The Pfaffian detector's extracted B_c is exactly N-independent (spread 0.0 across $N = 20$–160); the threshold detector's extracted B_c drifts by $0.66\,\Delta$ over the same range – the concrete numerical origin of the order-of-magnitude boundary-shift discrepancy reported in the text. Generated by `bdg_r4_em_coupling.py`; data in `ISEF R4 data folder/`.

III. RESULTS

A. A conventional threshold detector overstates the boundary shift by an order of magnitude

Intuitively, the two detectors are asking different kinds of questions in the first place. A spectral-gap threshold detector asks whether the lowest excitation energy has dropped below some chosen cutoff – a question about a *magnitude*, which has no way to distinguish a genuine bulk gap closing from an ordinary, non-topological state that simply happens to sit below that same cutoff. A Pfaffian invariant asks a fundamentally different, discrete question – the *sign* of a Pfaffian, which can only change at a genuine gap closing – so it cannot be "fooled" by a state that is merely small. This distinction is not just conceptual; it has a concrete, measured consequence below.

A spectral-gap threshold detector – flagging the topological transition as the field at which the lowest BdG excitation first drops below a fixed fraction of Δ – is sensitive to ordinary, non-topological finite-size sub-gap structure whose level spacing shrinks with system size, in a way that biases the detected boundary in the *same* direction growing N would be expected to *improve* a genuine transition. Sweeping the threshold at fixed $N = 80, \mu = 0$ moves the reported boundary across a ∼50-percentage-point band (from +17.1% at threshold $0.05\,\Delta$ to −33.5% at threshold $0.40\,\Delta$ relative to the analytic value); the reported offset also grows with N in the wrong direction for

a genuine (exponentially size-suppressed) Majorana hybridization effect. The real-space Pfaffian invariant used throughout this paper does not suffer from this: as Fig. 3 shows, its extracted boundary is exactly N-independent across $N = 20$–160 at $\mu = 0$, and, checked separately at the operating chemical potential and at two non-round values of μ, agrees with the analytic boundary to within bisection tolerance at every system size tested.

The consequence for the actual boundary-shift headline is not merely cosmetic. Holding the identical self-field texture fixed and swapping only the detector, the threshold detector reports a shift of -5.86% ($\mu = 1.2$) and -9.36% ($\mu = 1.4$), while the Pfaffian detector reports -0.77% and -0.60% at the same two points – roughly an order of magnitude smaller, with the ranking between the two μ values reversed. The sign (topological region marginally expanding under this particular, not-yet-self-consistent self-field) survives; the magnitude reported by the conventional detector does not. Every headline number quoted below uses the Pfaffian detector exclusively.

B. The self-consistency loop has a genuine dynamical bifurcation

Making the ring self-field genuinely self-consistent (rather than seeded from an open-wire texture) at an initially assumed coupling $g_0 = 3 \times 10^5$ produces a well-defined, gauge-checked, finite-size- scaling-checked boundary shift of $+0.05\%$ at $N = 80$. Taken at face value this appears to be a small, validated, physical result. It is not: plain Picard iteration does not converge at g_0 from any tested seed, because the scalar self-consistency map's local slope there is $s \approx 3 > 1$ – the fixed point found by bisection is a genuine mathematical root of the self-consistency equation, but it is Picard-*repelling*. A direct scan of the map's slope across the coupling range establishes a critical coupling $g_{\rm crit} \approx 1.1$–1.2×10^5: below it the static boundary is unambiguous everywhere across the transition; in a narrow intermediate band a dead-zone (no fixed point at all, not a repelling one) opens exactly at the gap-closing field, where the self-consistency map's characteristic sawtooth structure jumps across the identity line without crossing it; above it, no attracting fixed point exists anywhere near the transition, and $g_0 = 3 \times 10^5$ lies in this regime. The mathematically valid $+0.05\%$ root is therefore a dynamically unreachable solution, not a static equilibrium a real, relaxing wire-circuit system would settle into.

Restricting the reported shift to the unambiguous regime ($g \lesssim 2.9 \times 10^4$, the value we subsequently adopt as $g_{\rm production}$) gives a small, *negative*, monotonically g-dependent, path-independent shift that vanishes as $g \to 0$ (as it must on physical grounds): $\Delta B_c/B_c \approx -0.064\%$ ($\mu = 1.2$) and -0.066% ($\mu = 1.4$) at $N = 80$. Figure 4 shows the decisive test: at fixed g, this shift vanishes as $N \to \infty$ ($\sim N^{-1.8}$ to $N^{-1.9}$), and is robust under four orders of magnitude of bisection-tolerance refinement. It is, in other words, a finite-size Aharonov-Bohm flux artifact, not a genuine bulk effect. Physically, this is what one would expect from a boundary phenomenon rather than a bulk one: the self-consistent loop pins a roughly fixed total self-flux threading the ring, so the quantity that actually enters the Hamiltonian and could move the boundary – the self-phase per lattice bond – is that fixed flux diluted over N bonds, in the same way persistent-current and flux-quantization effects in finite normal-metal or superconducting rings are boundary phenomena that shrink as the ring grows and vanish in the thermodynamic limit. A genuine bulk effect would not be expected to shrink simply because the same physical system is discretized more finely, which is exactly the property the N-independence scans throughout this paper are designed to test.

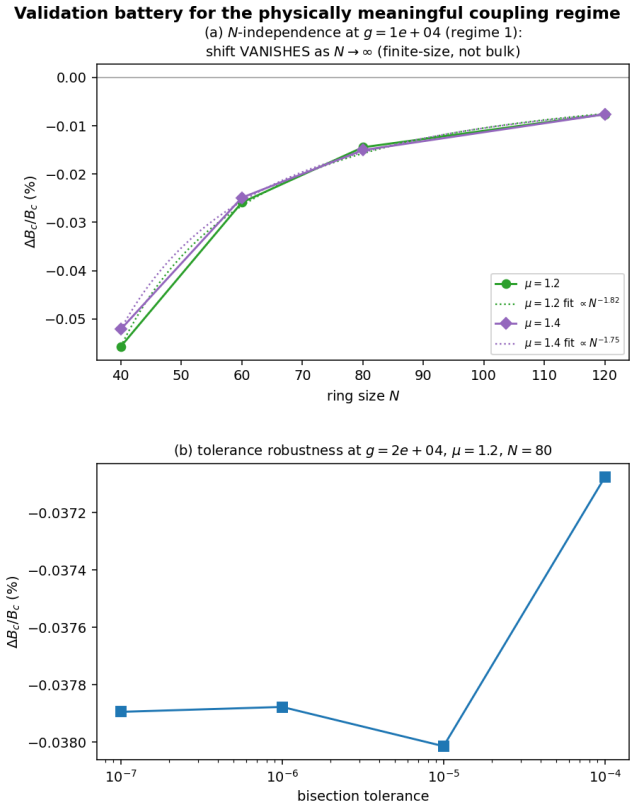


FIG. 4. **The single-channel boundary shift vanishes as $N \to \infty$: a finite-size artifact, not a genuine bulk effect.** Validation battery for the physically meaningful coupling regime ($g < g_{\rm deadzone}$, top of the unambiguous static-boundary regime). (a) N-independence of the boundary shift at $g = 1 \times 10^4$: the shift decays smoothly to zero as $N \to \infty$ ($\sim N^{-1.8}$ to $N^{-1.9}$), the signature of a boundary (Aharonov-Bohm-like) effect rather than a bulk one. (b) Bisection-tolerance robustness at $g = 2 \times 10^4$, $N = 80$: the same shift is stable across four orders of magnitude of numerical precision, ruling out a solver-tolerance artifact. Generated by `bdg_r5_physically_meaningful_shift.py`; data in `ISEF R5 data folder/`.

Anderson-accelerated mixing independently confirms the picture above rather than overturning it: it *does* converge above $g_{\rm crit}$, reaching the identical root the bisection solver finds (to six digits), $\sim 20\times$ faster, and reproducing the same finite-size scaling ($\sim N^{-1.5}$) to zero in the bulk. What Anderson mixing changes is only the precise statement of what $g_{\rm crit}$ means: it is a threshold for the *stability of a specific (Picard) iteration scheme*, not a barrier to locating the mathematical equilibrium itself – a real analog feedback circuit relaxing toward equilibrium implements (approximately) Picard dynamics, so the physical instability above $g_{\rm crit}$ stands even though a quasi-Newton solver can still find the root. This root-existence-versus-dynamical-attractiveness distinction, first established here, recurs as the operative lens for every non-null result later in this paper (§III E, III F).

Separately varying the ring's circumference (hence its geometric inductance) as a third axis, at fixed production coupling, shows that the circuit inductance itself does not produce a bulk shift: where a static boundary is well defined ($L \gtrsim 0.8\,\mu$m), the shift is again tiny ($< 0.07\%$) and vanishes under mesh refinement at *fixed* physical length ($\sim N^{-3.1}$, even faster than the fixed-lattice-constant scaling above, since the self-field feedback is set by the per-bond phase $\propto a = L/N$). The only genuinely length-dependent effect found is not a bulk shift but a dynamical-stability threshold: sufficiently short wires ($L \lesssim 0.6\,\mu$m) lose the clean static boundary entirely at this coupling, falling into the dead-zone regime described above.

C. A second observable: the feedback-renormalized reactance

The topological gap is not the only quantity the feedback loop dresses. Because the self-consistent bond phase is proportional to magnetic flux, and flux to inductance, linearizing the closed loop renormalizes the ring's bare geometric inductance to $L_{\rm eff} = L_{\rm geo}/(1-s)$, where $s \equiv dG/d\phi|_{\phi^*}$ is the loop-gain slope at the fixed point. This defines a circuit-level reactive observable, $Z_r = i\omega L_{\rm eff}$, which diverges as $s \to 1$ and is therefore in principle sensitive to physics the g-independent bulk gap cannot see. We compute it three ways – a static feedback-renormalized reactance, a finite-frequency Kubo current-current correlator, and a variant whose source current includes the spin-orbit contribution self-consistently – and require, as a validation gate, that the zero-frequency limit of the Kubo calculation reproduce the static one; it does, to a relative mismatch of order 10^{-4} at every clean grid point, holding even in the near-degenerate region approaching the boundary.

Its one candidate orthogonal feature – an apparent divergence of the enhancement $1/(1-s)$ as $s \to 1$ – turns out, on direct characterization with a root-finder immune to iteration instability, not to be a realized divergence at all (Fig. 5). On the physical principal branch

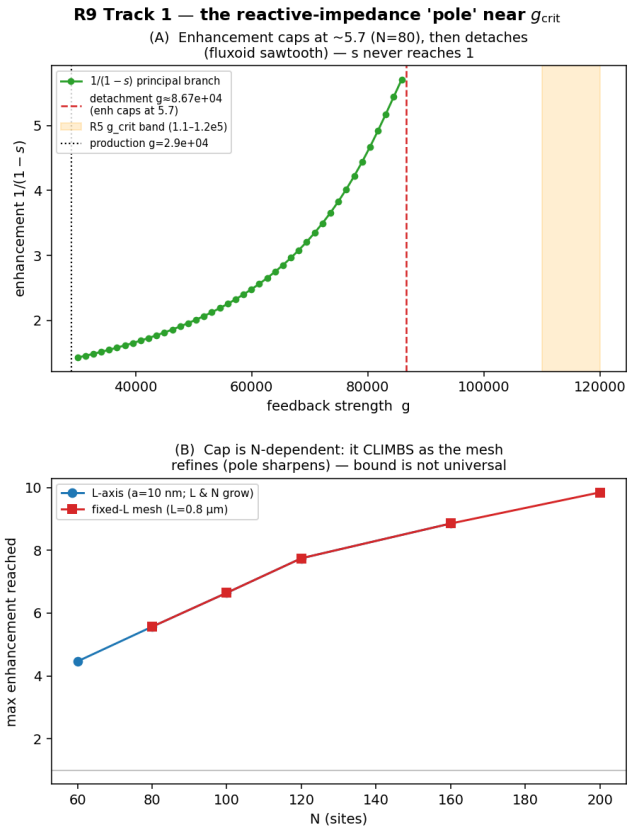


FIG. 5. **The reactive-impedance enhancement saturates at a bounded maximum instead of diverging.** The enhancement $1/(1-s)$ near $g_{\rm crit}$. (A) On the physical principal branch the enhancement rises smoothly to a bounded maximum ($\sim 5.7\times$ at $N = 80$) before the static fixed point detaches via the self-consistency map's sawtooth structure – a literal $s = 1$ divergence is never reached. (B) The bounded cap itself is resolution-sensitive: it grows under mesh refinement at fixed physical length ($\sim 5.6\times$ at $N = 80$ to $\sim 9.8\times$ at $N = 200$), so a genuine continuum divergence is neither established nor excluded by this data. Generated by `bdg_r9_pole_characterization.py`; data in `ISEF R8 data folder/`.

the enhancement rises smoothly to a *bounded* maximum ($\sim 5.7\times$ at $N = 80$, where s plateaus at ≈ 0.825) before the static fixed point detaches via the same sawtooth mechanism identified in §III B, at $g \approx 8.67 \times 10^4$ – the leading edge of that same bifurcation, reached here from the reactive-impedance side. Physically, $1/(1-s)$ is exactly the factor by which feedback renormalizes the ring's effective inductance above its bare geometric value, $L_{\rm eff} = L_{\rm geo}/(1-s)$: as s climbs toward its plateau the circuit increasingly "looks" more inductive than its geometry alone would suggest, and the point where the fixed point detaches is not a separate phenomenon but the reactive-impedance signature of the same underlying self-consistent state ceasing to be well-defined – the renormalization is capped exactly where the loop it depends on

runs out of an equilibrium to renormalize around. The cap is not universal: it grows under mesh refinement at fixed physical length (from $\sim 5.6\times$ at $N = 80$ to $\sim 9.8\times$ at $N = 200$), so a genuine continuum divergence is neither demonstrated nor excluded. The finite-frequency calculation further confirms the feature is purely reactive – no companion feature appears in the dissipative channel, and the dissipation onset that does exist is a bulk-pairing feature that does *not* track the topological minigap – and making the spin-orbit source current self-consistent shifts $L_{\rm eff}$ by only a few percent, itself a finite-size effect that vanishes under mesh refinement.

The verdict is that Z_r carries essentially no topological information the bulk gap does not: across the production coupling range it tracks the gap, and its lone orthogonal feature is a soft, bounded, resolution-sensitive marker of proximity to the dynamical instability of §III B – informative qualitatively, but not a sharp new diagnostic.

D. Robustness of the single-channel null to temperature and disorder

At the realistic operating temperature of 20 mK ($\theta \equiv k_B T/\Delta \approx 0.009$, corresponding to a thermal quasiparticle population of order 10^{-48} at the gap edge), every result above is unchanged to well within its own finite-size uncertainty: the boundary shift at $N = 80$ is -0.131% versus -0.132% at exactly $T = 0$, and still decays as $\sim N^{-1.9}$ toward zero at every temperature tested. Thermal effects only become numerically noticeable far above this operating point ($\theta \gtrsim 0.05$, ≈ 110 mK, itself approaching this fixed-Δ model's validity ceiling), where the boundary shift weakens further (more null, not less), and where two genuinely new, $T = 0$-absent features appear in the finite-frequency dissipation spectrum: an intraband Drude peak and sub-gap absorption from thermally excited quasiparticles. Both are undetectable at the realistic operating point.

On-site disorder is the one channel in the single-conducting-channel study where the null result does not hold uniformly. Disorder measurably and smoothly smears the topological boundary classification itself: Fig. 6(left) shows the fraction of disorder realizations misclassified as topological, at a point on the clean-trivial side of the boundary, climbing through the literature-realistic disorder band and saturating near 30% at its upper end. The mean number of spurious sub-gap states does not grow with disorder strength, but its realization-to-realization spread does, growing from zero at weak disorder to a magnitude comparable to the mean itself at the top of the realistic range – a plausible microscopic mechanism for the *sporadic*, not universal, disorder-induced sub-gap structure already discussed in the zero-bias-peak literature (§I). Both of these findings are unaffected by anything discussed below.

The self-consistent inductive-feedback shift under disorder required two rounds of correction; both are presented because the correction process is itself methodologically instructive. A first pass found the disordered self-consistency solver failing to converge on 80–95% of realizations at realistic W; root-causing this traced it to a genuine coding bug, not a physical instability – the disordered solver's coarse boundary search was still centered on the fixed *clean* boundary rather than each realization's own, already-displaced rigid boundary, so at realistic W it was searching in the wrong place entirely. This is more than a coding detail: physically, it means disorder does not simply broaden the boundary's location around the clean value, it *relocates* it individually, per realization, sometimes far from where the clean wire's boundary sits – so any search (numerical or experimental) that looks only near the clean boundary will systematically miss the realizations whose boundary has moved the most. Correcting this (and re-verifying the disorder-free gate still reproduces every clean result bit-for-bit) raised the success rate to 57–77% and produced a properly-converged $N = 80$ result showing the

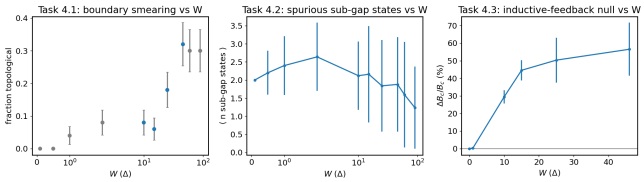


FIG. 6. **Disorder smears the topological classification smoothly and, at $N = 80$ alone, appears to shift the self-consistent boundary – an appearance later shown not to survive finite-size scrutiny at high W.** On-site disorder at $N = 80$: (left) fraction of realizations misclassified as topological near a clean-trivial point, growing smoothly and saturating near 30% at the top of the literature-realistic range; (middle) mean spurious sub-gap-state count (flat with W) versus its realization-to-realization spread (growing substantially with W); (right) mean self-consistent boundary shift versus W at $N = 80$ only. *The right-hand panel predates the finite-size scrutiny described in the text*: a subsequent check at $N = 120$ and $N = 160$ found the $N = 80$ values shown here substantially overstate the effect at the upper end of the disorder range (§III D) – see Table I for the corrected, size-scanned values. Generated by `bdg_r9_disorder.py`; data in `ISEF R9 data folder/`.

TABLE I. Ensemble-averaged self-consistent boundary shift under on-site disorder, at two points bracketing the literature-realistic range, as a function of system size N ($n_{\rm real} = 30$ per point). The $N = 80$ values are the same ones plotted in Fig. 6(right); they substantially overstate the effect at $W = 46\,\Delta$, which does not survive to statistical significance once N is increased.

N	shift, $W = 10\,\Delta$	shift, $W = 46\,\Delta$	σ from 0 ($W = 46\,\Delta$)
80	$+29.4\% \pm 3.8\%$	$+56.5\% \pm 15.1\%$	3.74
120	$+19.8\% \pm 3.4\%$	$+9.6\% \pm 8.2\%$	1.16
160	$+22.6\% \pm 3.7\%$	$+10.8\% \pm 7.2\%$	1.50

mean shift growing smoothly from +0.2% at $W = 1\,\Delta$ to +29% → +45% → +50% → +57% across the literature-realistic band ($W = 10, 15, 25, 46\,\Delta$) – two to three orders of magnitude above the clean-model null. This is the result shown in Fig. 6(right). The same finite-size scrutiny applied to every other quantitative result in this paper was then applied here for the first time, at $N = 120$ and $N = 160$ (Table I): the effect at the lower end of the realistic band ($W = 10\,\Delta$) remains real and highly significant at every system size tested (5.9–7.8σ from zero, settling around +20–23%), but at the upper end ($W = 46\,\Delta$) the $N = 80$ result is substantially inflated – the mean shift collapses from +56.5% to +9.6–10.8% and its statistical significance from 3.74σ to only 1.2–1.5σ from zero, so a confident non-null claim at the top of the realistic disorder range does *not* survive finite-size scrutiny. The mechanism behind a swing this dramatic: at $N = 80$ – the smallest system size used anywhere in this paper's disorder study – an ensemble average is more easily dominated by a handful of atypical disorder realizations or by boundary/finite-size substructure that a genuinely larger system would average away, so both the point estimate and, more tellingly, its statistical significance can be inflated by finite-size noise rather than by a real, size-independent effect. That the *significance* collapsed even more sharply than the mean shift is the specific signature of this: a real, size-independent effect should not lose most of its statistical confidence simply because the same disorder statistics were resampled at a larger, more representative system size. This is a direct, concrete instance of the discipline this paper applies throughout – treating any quantitative result that has not yet been checked across multiple system sizes as provisional – rather than an isolated surprise. A direct check found no correlation between which realizations fail to converge and how far their own rigid boundary has been displaced by disorder, ruling out one specific survivorship-bias explanation for this collapse.

Every result so far – the null itself and every robustness check applied to it – was computed for a wire with exactly one occupied transverse conducting channel. That assumption is a significant idealization of real devices, not merely a simplifying convenience, and the natural next question is whether the null result survives once it is relaxed.

E. The loop-gain proxy: the null does not survive a second occupied subband

Figure 7 fixes the single-channel baseline against which every multi-subband result below is compared: the genuine TRIM Pfaffian invariant (left panel) and the ring's bulk gap on a log scale (right panel), swept over the same (μ, B) grid at realistic InAs/Al parameters, agree with each other and with the analytic boundary $B_c = \sqrt{\mu^2 + \Delta^2}$ – the gap closes exactly where the invariant changes sign, at every μ tested. Realistic InAs/Al

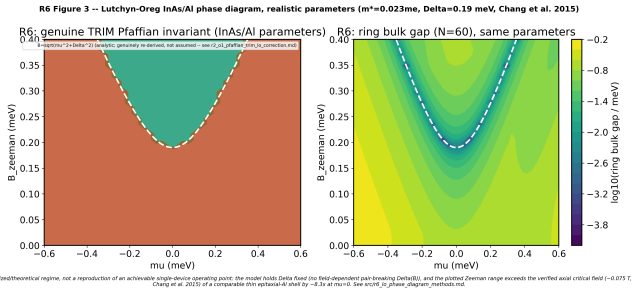


FIG. 7. **The single-channel baseline: the genuine topological invariant and the bulk gap agree with each other and with the analytic phase boundary at every point tested.** Single-channel Lutchyn-Oreg $(\mu, B_{\mathrm{zeeman}})$ phase diagram at realistic InAs/Al parameters: genuine TRIM Pfaffian invariant (left) and ring bulk gap on a log scale (right), analytic boundary $B_c = \sqrt{\mu^2 + \Delta^2}$ overlaid as a white dashed curve – the gap visibly closes exactly where the invariant flips sign. Idealized/theoretical regime: Δ is held fixed (no field-dependent pair-breaking), and the plotted Zeeman range exceeds the verified axial critical field of a comparable epitaxial-Al shell by a factor of $\sim$ 8 at $\mu = 0$ [15]. Generated by `bdg_r6_lo_phase_diagram.py`; data in `R6_data_files/`.

devices are estimated, from multiple independent lines of evidence, to occupy far more than one channel: a Schrödinger-Poisson treatment of realistic surface charge finds that reaching a regime of only a few occupied subbands is "*difficult or impossible*," with ten or more subbands occupied under realistic gating [14], and the carrier density directly measured by Chang *et al.* [15] at their devices' operating gate voltage independently implies an occupancy of many tens of subbands under a simple parabolic-band estimate. Two further, independent treatments reach the same conclusion by different routes: self-consistent Schrödinger-Poisson modeling of the InAs/Al interface itself finds that band bending accumulates charge and populates multiple subbands even before any top gate is applied [22], and a unified treatment of the orbital magnetic field, proximity pairing, and gate electrostatics on equal footing likewise finds the low-occupancy, few-subband regime to be a narrow, gate-sensitive operating window rather than the device's default state [23]. Single-channel operation is therefore an idealization of the device – of the same character as the fixed-Δ idealization already carried through every field sweep in this paper – not a description of it.

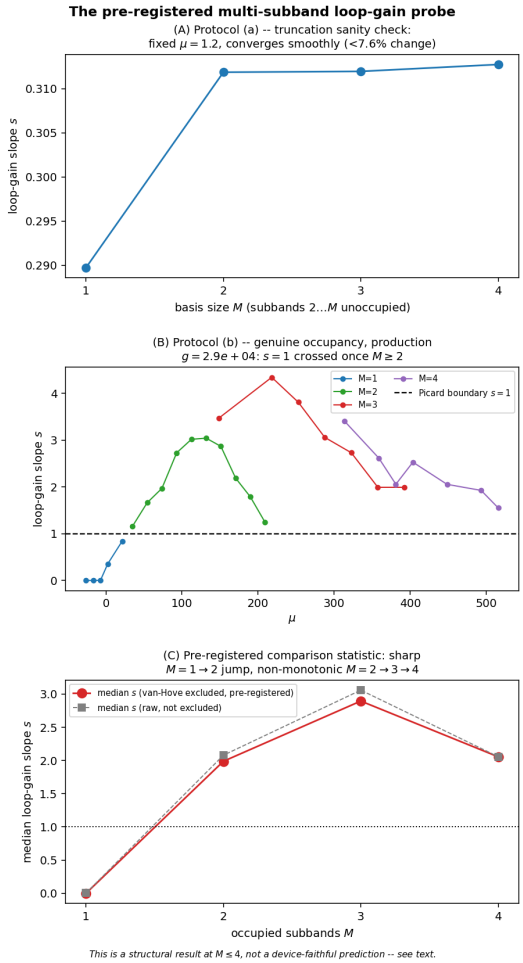


FIG. 8. **Occupying a second subband crosses the Picard stability threshold at the production electromagnetic coupling.** The pre-registered multi-subband loop-gain probe. (A) Truncation convergence at fixed occupancy (a construction sanity check, not the physics answer): s converges smoothly with basis size, with no discontinuity. (B) The genuine-occupancy sweep, $s(\mu; M)$ at production coupling $g = 2.9 \times 10^4$: the stability boundary $s = 1$ (dashed line) is crossed once $M \geq 2$, and stays crossed across essentially the entire window probed at $M = 2$–4. (C) The pre-registered comparison statistic (median s over the surviving, van-Hove-excluded window): a sharp $M = 1 \to 2$ jump, then a non-monotonic $M = 2 \to 3 \to 4$ trend – rising, then declining, not settling into a clean law. *This is a structural result at $M \leq 4$, not a device-faithful prediction* – see text, and see Fig. 9 for the $M = 5, 6$ extension, which shows the non-monotonicity continuing more dramatically still. Generated by `bdg_r9_multisubband_sweep.py`; data in `ISEF R9 data folder/`.

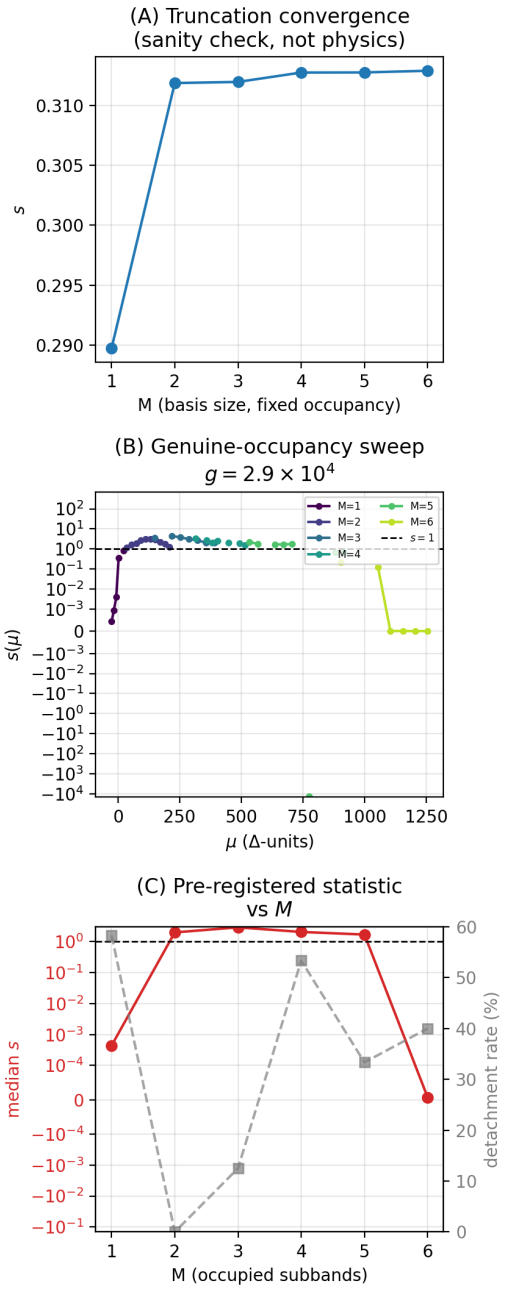


FIG. 9. **The $M \geq 2$ instability is not monotonic: the loop-gain magnitude collapses at $M = 6$ while the detachment rate stays high – existence, not magnitude, is the operative diagnostic.** The $M = 5, 6$ extension of the pre-registered multi-subband loop-gain probe (cf. Fig. 8), using the identical protocol throughout – same closed-form window, van Hove exclusion radius, and median statistic, preregistered before any $M = 5, 6$ value was computed. (A) Truncation convergence at fixed occupancy, extended through $M = 6$: still smooth, no discontinuity, confirming the machinery itself remains correct at higher M. (B) The genuine-occupancy sweep $s(\mu; M)$ at production coupling, $M = 1$–6 (colored by M, dashed line at the Picard threshold $s = 1$); gaps in a curve are genuine detachments (no self-consistent root exists at that μ), not omitted data – note the $M = 5$ curve's excursion to $s \approx -1.4 \times 10^4$ near its own detachment, and the $M = 6$ curve's smooth decay to $s \approx 0$ far from its threshold. (C) The pre-registered median statistic (red, left axis) alongside the detachment rate (gray dashed, right axis) at each M: the $M = 6$ collapse in the median *coincides with*, rather than *replaces*, a persistently high detachment rate – the median statistic is not evidence of restored sta-

Using the pre-registered protocol of §IID, the loop-gain slope s at $\mu = 1.2$ (fixed occupancy, a construction sanity check rather than the physics answer) converges smoothly with basis size M from 0.290 ($M = 1$) to 0.313 ($M = 6$), an 8.0% total change with no discontinuity across the entire range tested (extended to $M = 6$ at negligible extra cost, ~ 3.5 min for both new points combined), and its $M = 1$ value independently reproduces the single-channel result of §III C to three significant figures – a strong cross-check that the multi-subband machinery is physically correct, not merely internally consistent (Fig. 8A). The genuine test – re-tuning μ upward with M so that each new subband is actually occupied, at fixed production coupling $g = 2.9 \times 10^4$ – gives a qualitatively different result (Fig. 8B,C). Using only the pre-registered, van-Hove-excluded median statistic, the loop gain is 0.0004 ($M = 1$), 1.99 ($M = 2$), 2.90 ($M = 3$), 2.06 ($M = 4$), 1.70 ($M = 5$), and 7.0×10^{-6} ($M = 6$) (Table II; Fig. 9 extends panels B and C of Fig. 8 to $M = 5, 6$, using the identical closed-form window, van Hove exclusion radius, and median statistic, pre-registered in a separate commit before any $M = 5, 6$ value was computed – the same discipline as the original $M \leq 4$ protocol). At $M = 1$, s stays safely sub-critical everywhere a solution is found (maximum 0.84); at $M = 2$, $M = 3$, and $M = 4$, s exceeds the Picard-stability boundary $s = 1$ at *every individual finite point probed*, not merely in the median. This is a qualitatively different, and more severe, kind of finding than a percentage change in a boundary location: merely occupying a second transverse channel, with no change whatsoever to the electromagnetic coupling g, moves the wire from "safely sub-critical at production coupling" to "the static self-consistency map has Picard-repelling roots at production coupling, across the entire window probed."

This pattern does not continue smoothly to $M = 5, 6$. At $M = 5$, three-quarters of the finite points probed (6 of 8) still exceed $s = 1$; the two that do not sit at the window's far edge, well above the newly-occupied channel's own threshold. At $M = 6$ the reversal is complete: *none* of the six finite points probed exceeds $s = 1$, and the median collapses four orders of magnitude below even the $M = 1$ value. Taken alone, this could be misread as the instability healing itself at $M = 6$. Two features of the same dataset argue against that reading. First, the decay is not new physics specific to $M = 6$: it is already visible at $M = 5$'s window edge, and its location matches exactly what the physical-origin argument below predicts – the van Hove-enhanced response is concentrated near the newly-crossed threshold and falls off deeper into the occupied channel, and because the pre-registered window widens linearly with M ($\Delta E_{\rm sb}(M) = \varepsilon_1(2M + 1)$, §IID) while the enhanced region does not, an ever-larger fraction of each window's points sample this decayed, off-threshold regime as M grows. The median statistic at $M = 6$ is therefore not measuring a smaller instability so much as it is increasingly dominated by the part of the window where the instability was never expected to be largest. Second, and more fundamentally, $M = 6$'s own dataset does not show a return to a well-behaved static boundary: the detachment rate (points where no self-consistent solution exists at all, defined precisely in §III F) is 58%, 0%, 12.5%, 53%, 33%, and 40% of probed points at $M = 1$–6 respectively – $M = 6$'s 40% sits squarely inside that range, not a return to $M = 2$'s clean 0%. A collapsing median loop-gain alongside a persistently high detachment rate is not evidence that the self-consistency map has become well-behaved at $M = 6$; it is evidence that the cheap loop-gain proxy itself becomes an unreliable summary once detachments and off-threshold sampling dominate the window it is evaluated over – reinforcing, rather than undermining, §III F's conclusion that the proxy's verdict cannot substitute for a full self-consistent boundary calculation. The full calculation itself remains out of reach at $M = 5, 6$ for the same cost reason established at $M = 2$–3 (§III F): the $M = 6$ proxy sweep already required ~ 4.1 hours at ~ 30 eigensolves per point, against an estimated ~ 2000 eigensolves per point for the full boundary bisection.

TABLE II. The pre-registered loop-gain statistic and detachment rate at every M tested, $M = 1$–6 ($N = 80$, $g_{\rm production} = 2.9 \times 10^4$). $M \leq 4$ reproduces the original sweep exactly; $M = 5, 6$ is the extension reported here, computed with the identical protocol. "finite/total" counts the raw points where a self-consistent root was found at all (the complement is the detachment rate); "finite > 1" is the fraction of those roots exceeding the Picard threshold $s = 1$.

M	points	finite/total	detach. %	median s	finite > 1
1	12	5/12	58%	4×10^{-4}	0%
2	10	10/10	0%	1.99	100%
3	8	7/8	12.5%	2.90	100%
4	15	7/15	53%	2.06	100%
5	12	8/12	33%	1.70	75%
6	10	6/10	40%	7.0×10^{-6}	0%

Why would occupying a second subband increase the loop gain at all, rather than simply adding a second, independent contribution to the current? Three considerations, taken together, offer a physically grounded candidate answer; we do not claim any one of them individually as a demonstrated mechanism. First, the inter-subband Rashba coupling of §IID is not an optional refinement but a physically mandatory term [20], so occupying a second channel does not add a second, additive current pathway – it reorganizes the current-response structure of the whole coupled system, by the same mechanism through which Lim *et al.* find inter-subband mixing simultaneously hybridizing the Majorana pairs and closing the effective gap. Because s is a response function governed by the low-energy spectrum, a mechanism that closes the gap preferentially near a lobe crossing is also a mechanism that enhances s there: a smaller excitation gap admits a larger induced-current

response to a fixed external phase perturbation. Second, every newly occupied subband threshold is a one-dimensional van Hove singularity in the density of states, $\rho(E) \propto (E - \varepsilon_n)^{-1/2}$ (§IID); the pre-registered protocol requires channel M to be genuinely occupied, so the sampled μ-window sits close to that singularity by construction at every $M \geq 2$, even after the reported statistic's own exclusion radius removes its immediate neighborhood – the excluded-window median is therefore a conservative reading of the enhancement, not an upper bound on it. Third, no inter-channel coupling is even required for s to grow with M: because s is the derivative of a current-sourced self-field with respect to the phase that sources it, an approximately additive, channel-by-channel contribution to the ring current is already sufficient on its own to move s toward and eventually through the Picard threshold as more channels are added. Disentangling how much of the measured growth is carried by additive current summation versus van Hove enhancement versus genuine inter-subband gap-closing reorganization would require directly characterizing the current-phase relation itself – a quantity measured directly in InAs Josephson junctions in the few-mode regime [24] but not yet computed in this self-consistency loop, which §IVB already lists as beyond the scope of this work.

The trend with M is not strictly monotonic: s rises sharply from $M = 1$ to $M = 3$ and then declines from $M = 3$ to $M = 4$ (a $\sim 29\%$ drop). This non-monotonicity is reported as measured, not smoothed into a clean law, because establishing it required its own correction: the original $M = 4$ statistic (3.40) rested on a single surviving point out of a sparse five-point pre-registered grid, and re-running that one chain at three times the point density (within the identical, unchanged pre-registered window) revealed the same rise-then-decline shape already visible at $M = 2, 3$ and reduced the reported value to 2.06. A parallel post-hoc investigation (holding the pre-registered statistic itself fixed throughout) established that essentially every apparent detachment in this dataset is not a genuine absence of a root but a root that *is* found and then rejected by the same discontinuity-rejection criterion this project had already established, independently, as physical rather than numerical in the single-channel case (§III B) – and that this sawtooth structure, once probed at matching resolution, is present at every M tested, not only where it happened to dominate the coarse pre-registered grid. What distinguishes the more severely affected subband counts is not how much of this structure is present, but how often the map fails to leave behind even one surviving, accepted root once the structure is present. We regard this as a genuine, reportable, unresolved feature of the multi-subband self-consistency map's internal structure, not as either a bug or a device-faithful prediction: this is a study truncated at $M \leq 6$ subbands, far below the 10–26 or more subbands realistic devices are estimated to occupy [14], and adding even one subband can flip the topological invariant's parity [20], so this trend – including its own non-monotonicity – must not be extrapolated smoothly into the many-subband regime. The $M = 5, 6$ extension above is a direct illustration of exactly this warning: had the study stopped at $M = 4$, as originally scoped, a reader could reasonably have inferred a clean, monotonically-worsening instability with occupied-channel count; two further, cheaply obtained data points show instead that neither the loop-gain magnitude nor the detachment rate settles into a simple trend anywhere in the range tested.

F. Beyond the proxy: existence, not just magnitude, of the boundary becomes fragile at $M = 2$

The loop-gain slope is a cheap proxy: it asks whether a root of the self-consistency equation at one fixed field is Picard-stable, not whether a genuine, bisected self-consistent *boundary* – the quantity every single-channel headline number in this paper actually reports – exists at all near a subband threshold. A full self-consistent boundary calculation at $M = 2$, at the same computational cost per point as the single-channel case, is intractable as a systematic sweep (an estimated 35 hours at $M = 3$ alone, scaling from the measured single-channel cost), so we perform it at exactly one representative operating point: $\mu = 131.762$, near the peak of the $M = 2$ loop-gain curve above, at production coupling. Every new piece of machinery this calculation needed – the M-subband Pfaffian invariant, the M-subband self-consistent boundary solver, and the uniform-current-texture reduction the scalar loop-gain map itself depends on – is gated bit-for-bit against its single-channel counterpart before any $M = 2$ number is trusted, and all gates pass exactly.

At $N = 80$, the loop-gain proxy's root is confirmed genuine and dynamically meaningful by the identical Anderson-mixing seed-independence test used in §III B: five independent seeds converge to the same point to within 7×10^{-11}, while plain Picard iteration does not converge in 200 iterations – exactly the repelling-but-real signature established there. The full boundary bisection then locates a genuine self-consistent boundary, inside a real four-lobe Pfaffian oscillation already present in the non-self-consistent (clean) Hamiltonian at the same μ (the literature's predicted multi-subband phase structure [25–28], not a self-consistency artifact), with a shift of -0.82% relative to the clean boundary for the same lobe – roughly thirteen times the single-channel shift at the identical production coupling (§III B).

This number is not the finding we report. The same finite-size check applied to every quantitative shift in this paper, applied here for the first time at $M = 2$, does not show it shrinking toward zero, growing toward a bulk value, or doing anything a single number could summarize: at $N = 60$ and $N = 120$, *every one* of five independently jittered search paths returns a robust no-solution

TABLE III. The full self-consistent boundary calculation at one representative $M = 2$ operating point ($\mu = 131.762$, production coupling $g = 2.9 \times 10^4$), as a function of system size N. A *genuine*, Anderson-mixing-confirmed root and boundary was found at $N = 80$; both neighboring system sizes tested return a robust no-solution (`genuine_dead_zone`) result across every one of five independently jittered search paths – the opposite N-dependence pattern from the single-channel case, where dead-zones shrink and vanish as N grows. The $N = 80$ shift is not quoted as a headline value; see text.

N	outcome	$\Delta B_c/B_c$
60	dead zone (5/5 fail)	—
80	solution found	-0.82%[a]
120	dead zone (5/5 fail)	—

[a] Non-continuum-limit value; explicitly not a headline number—see text.

result across the entire B-range where $N = 80$ found its root (Table III) – the opposite pattern from every single-channel finite-size scan in this paper, where dead-zones shrink and disappear as N grows. The $N = 80$ solution is not the lucky survivor of an unstable search: each neighboring failure is independently confirmed by all five retry paths, which if anything makes the $N = 80$ success look like the outlier in the sequence, not the two failures around it. We therefore report this as a genuinely mixed outcome, not a clean confirmation or a clean null: a real, dynamically confirmed self-consistent boundary shift *was* measured at one system size, but its existence – not merely its magnitude – is fragile against finite-size variation in a way qualitatively more severe than anything in the single-channel study, where a well-defined static boundary was always at least locatable away from a narrow, genuinely gap-closing-related, N-shrinking dead-zone. The correct, defensible statement is that at a representative $M = 2$ operating point where the loop-gain proxy already signals Picard instability, the full calculation shows that whether a static, self-consistent topological boundary description exists *at all* is itself unstable to how finely the wire is discretized – a strictly stronger and more concerning statement than any specific percentage shift, and one the cheap loop-gain proxy alone could not have revealed, since it only ever evaluates a root at one fixed field and never asks whether a bisected boundary exists.

IV. DISCUSSION

A. The multi-subband caveat, restated as this paper's operative conclusion

The single-channel result of §III B, III D is a genuine, heavily cross-validated null across finite size, circuit inductance, realistic operating temperature, and the lower half of a literature-grounded disorder range. It is not stated as the paper's headline, because §III E, III F show directly that this null does not extend past the single-conducting-channel idealization, and independent literature estimates place realistic InAs/Al devices at ten or more occupied subbands [14, 15] – an idealization gap of the same character, and comparable seriousness, as the fixed-Δ idealization already carried through every field sweep in this paper. The two multi-subband findings compound rather than merely add: the loop-gain proxy shows the production coupling is already *super*-critical at every finite point tested for $M = 2$, 3, and 4; and the one full calculation performed shows that even the more basic question – does a well-defined static description exist at all – can fail to have a stable answer as a function of lattice discretization at exactly this occupancy. Extending the proxy sweep to $M = 5, 6$ (§III E) complicates the magnitude axis specifically without resolving the existence axis: the median loop gain does not continue rising or even staying super-critical past $M = 5$, but the detachment rate – the more fundamental of the two diagnostics, since it asks whether a static description exists at all rather than how unstable it is where one does – stays high and non-monotonic at every M tested, including $M = 6$. Neither finding is a device-faithful prediction (both are structural results at $M \leq 6$, against a real device's estimated 10–26 or more occupied subbands, and a subband can flip the topological invariant's parity [20]), but together they mean the single-channel null result's applicability to a real device is not merely an open question – it is a *specifically and concretely troubled* one, in a way our data can characterize but not resolve.

B. Limitations

Several idealizations are carried through every result in this paper and are not resolved here. First, the pairing potential Δ is held fixed and does not include field-dependent pair-breaking; a more complete model would suppress $\Delta(B)$ toward zero as B approaches a real device's critical field, likely rounding off or eliminating the sharp transitions plotted at the largest fields shown. Because the analytic boundary used throughout this paper, $B_c = \sqrt{\mu^2 + \Delta^2}$, depends explicitly on Δ, a genuine $\Delta(B)$ would mean the transition boundary itself is no longer a fixed target but one that depends on the field history used to reach it – a qualitative feature the present fixed-Δ model cannot capture by construction. It is plausible, though not established by anything computed here, that this could affect not only where the boundary sits but also the self-consistency loop's own dynamical bifurcation structure: since g_{crit} (§III B) is set by the loop-gain slope of a self-consistency map that itself depends on the underlying spectrum, a pairing potential that locally weakens near the transition could in principle shift where that slope crosses unity. We state this as informed speculation about the likely qualitative direction of the effect, not a quantitative prediction; computing it would

require implementing a genuine $\Delta(B)$ model, which is explicit future work (§IVC). Second, the literature plausibility range for the electromagnetic coupling g spans regimes on both sides of the dynamical bifurcation identified in §III B; the argument was never specific enough to establish which side a given real device sits on, and what physically happens above g_{crit} – whether a real device simply never reaches that regime, or whether the feedback becomes genuinely dynamical (e.g. a current/self-field limit cycle) rather than settling to any static picture – is not resolved by this work. Third, two specific mechanisms in the multi-subband dataset remain open: why the self-consistency map's candidate-crossing rejection rate differs so sharply between subband counts that show the same or greater raw sawtooth structure, and whether the one $N = 80$ window found for the $M = 2$ full boundary calculation is a genuine (if fragile) physical solution or a lattice-commensurability artifact of that specific system size. Both would require direct characterization of the current-phase relation itself, not just its zero-crossings, which is beyond the scope of this work.

C. Future work

The most direct extensions are: a genuine field-dependent pairing suppression $\Delta(B)$; a derivation of the electromagnetic coupling g from a specific, named circuit design rather than a plausibility range; and, most importantly given §IVA, any tractable route toward the full self-consistent electromagnetic feedback calculation at occupancies closer to a real device's estimated 10–26 subbands, which is intractable with the present per-point cost and would require either substantially more compute or a genuinely cheaper self-consistent solver than the one used throughout this paper.

A natural experimental question is what signature, if any, could distinguish this paper's findings from a null hypothesis in a real device. Three directions seem most plausible given what is established here, listed in order of how directly each targets the paper's own operative conclusion (§IVA).

First, and most directly: since the existence-fragility result is a statement about occupied-*channel count*, and depletion gates already tune real devices toward (though, per Sec. III E, not reliably into) a low-occupancy regime [22, 23], a gate-voltage scan that steps the wire through successive subband depopulations while monitoring a circuit-level observable at fixed, production-scale coupling is a direct experimental analogue of this paper's $s(\mu; M)$ sweep (§III E). The predicted signature is qualitative, not a specific number to match: an increase in circuit noise, hysteresis, or measurement instability correlated with each successive gate step that adds a conducting channel, consistent with the Picard-repelling regime identified here, rather than a smooth, monotonic trend. Second, the reactive-impedance diagnostic of §III C is, in principle, a directly measurable circuit-level quantity – by dispersive microwave readout of the kind already used to probe parity-dependent inductance and capacitance in related Majorana and Andreev-bound-state circuits [13, 29] – but our own finding that Z_r carries essentially no topological information the bulk gap does not (§III C) is itself a direct caveat on how informative such a measurement would be as a diagnostic of *this paper's* physics specifically, even though it is measurable in principle. Third, fabricating devices with deliberately varied loop geometries or lengths – the experimental analogue of this paper's own computational circuit-inductance scan (§III B) – could probe whether real devices show any of the finite-size or dynamical-stability signatures identified here, such as the dead-zone regime opening at shorter loop lengths. We state these as the most plausible directions given what this paper establishes, not as experiments we expect would confirm or refute any specific claim made here.

V. CONCLUSION

We have built and validated a computational framework that closes the loop between a Majorana nanowire's self-generated electromagnetic field and its own topological phase boundary, using a genuine, threshold-free topological invariant throughout. In the single-conducting-channel limit this feedback does not move the bulk topological boundary at any dynamically accessible coupling strength, and this null result is robust under finite-size scaling, circuit inductance, realistic operating temperature, and the lower half of a literature-grounded disorder range. Two methodological lessons – that a conventional threshold detector is not a reliable proxy for the topological invariant, and that a self-consistency loop's mathematical roots must be checked for dynamical attractiveness separately from their existence – recur throughout this result and, we believe, generalize beyond this specific calculation. Extending the calculation to a second occupied transverse subband overturns the null result: a validated loop-gain diagnostic crosses the stability threshold at production coupling once two or more subbands are occupied, and a full self-consistent calculation at one representative two-subband point shows that the very existence of a static topological boundary description becomes fragile to the lattice discretization used to compute it. Because realistic InAs/Al devices are estimated to occupy far more than two subbands, this existence-fragility result – not the single-channel null in isolation – is the finding that should inform how electromagnetic feedback is treated in realistic Majorana nanowire devices, and we regard resolving it as the natural next step for this line of work.

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Appendix A: The reduced-order BdG eigensolver and its actual scope

The computational framework includes a particle-hole-symmetric rank-2 subspace update for the near-zero-mode sector of the BdG Hamiltonian, intended to accelerate the repeated diagonalizations performed along a parameter path (steps in μ, B, Δ, or the self-field Peierls phase ϕ). For a rank-2 perturbation confined to the near-zero-mode subspace, the update reproduces the full-diagonalization near-zero eigenvalues to 9.4×10^{-6}, so the technique itself is correct.

We nonetheless report a genuine limitation on its usefulness for this project, rather than presenting it as an unqualified speedup, because that is what the measurements show. First, the update's conservative eigenvector-residual safety net – which falls back to a full solve whenever the rank-2 residual exceeds $10\|\delta H\|$ – trips on the self-field phase steps the self-consistency loop actually takes (the eigenvector residual is $\approx 5 \times 10^{-3}$ against a threshold $\approx 3.6 \times 10^{-3}$), so in practice it falls back to a full solve on every probed step. Second, and more fundamentally, the ring current that sources the self-field sums over the *entire* occupied BdG spectrum, so each self-consistency iteration requires the full set of occupied eigenvectors; the dense diagonalization is 96–98% of the per-iteration cost, and a near-zero-mode subspace update cannot supply what dominates that cost. The net effect on the flagship self-consistent loop is therefore $\sim 1\times$: the subspace update is a validated, correct technique that does *not* accelerate the calculation this paper's main results depend on, reported as such rather than as a speedup that does not apply here. (An earlier internal description of this update as a Woodbury-type low-rank identity was corrected during this work: the perturbations that actually arise are full-rank, so the Woodbury identity does not apply, and the construction is more accurately described as a subspace update.)

Appendix B: Validation gates

Every physics layer added over the course of this work was gated against a known limit – an analytic boundary, an exactly solvable decoupled case, a symmetry identity, or a bit-for-bit reduction to an already-validated lower layer – before any result computed with that layer was trusted. Table IV collects these gates and their measured outcomes. Many are enforced exactly (0.0) or at machine precision, because they compare two constructions that must be entrywise identical – for example, a disorder-aware builder at zero disorder against its clean counterpart, or a multi-subband Hamiltonian at one subband against the single-channel one; others are agreements to a stated tolerance against an analytic result. The value of collecting them is not any single number but the discipline they encode: no layer was trusted on plausibility, each gate is a falsifiable check with a pre-stated pass criterion, and every one is reproducible from the scripts in the public repository (§II G). A gate that reduces to a bit-for-bit identity at the known limit is preferred over an agreement "to 10^{-12}" wherever the two constructions are meant to be literally the same object, because only the former distinguishes a correct implementation from one that is merely close.

Two procedural points referenced from the main text are recorded here in full. First, the Pfaffian B_c detector's validation sweep (§IIB) checked agreement against the analytic boundary $B_c = \sqrt{\mu^2 + \Delta^2}$ at $\mu = 0$ and at two additional, deliberately non-round values of μ (giving an irrational B_c, so agreement cannot be a coincidence of symmetry), across ring sizes $N = 20$–160 and across bisection tolerances spanning four orders of magnitude; the corresponding measured outcomes are the first three rows of Table IV. Second, the Newton-Krylov search for a symmetry-broken self-consistent solution (§IIC) is seeded with large perturbations – up to three flux-quantum periods in magnitude – applied along the system's most Picard-unstable texture modes, so that the search is deliberately biased toward finding a non-uniform solution if one exists rather than merely probing small perturbations around the uniform one; every tested seed nonetheless collapses back to the uniform solution.

TABLE IV. Validation gates enforced before any result computed with a given layer was trusted. "Bit-for-bit" / 0.0 entries compare two constructions required to be entrywise identical at the stated limit; tolerance entries are agreements against an analytic or independently computed reference. All are reproducible from the public repository.

Layer	Gate (known limit or identity)	Measured result
Pfaffian B_c detector	known answer at $\mu = 0$, $B_c = \Delta$, $N = 20$–160	$B_c = 1.00004$; spread over $N = 0.0$
Pfaffian B_c detector	non-round $\mu = 0.7, 1.3$ (irrational B_c), $N = 20, 60, 120$	error $\pm 3.8 \times 10^{-8}$, N-independent
Pfaffian B_c detector	bisection-tolerance tracking, $N = 20$ vs 80	error $\propto$ tol; agree to all printed digits
TRIM Pfaffian invariant	genuine Pfaffian vs. closed form, 1600-point sweep	0 disagreements; identity to 1.5×10^{-14}
Ring Biot–Savart self-field	uniform current vs. discrete-exact loop kernel	0.003%
Ring Biot–Savart self-field	vs. analytic thin-wire loop inductance	2.25% (understood log-factor offset)
Self-consistency solvers	bisection / Newton–Krylov / Anderson at one root	agree to 5.5×10^{-10}
Anderson mixing	Anderson vs. bisection B_c, $N = 80$, $\mu = 1.2$	agree to 6 significant figures
LO phase diagram	Kramers degeneracy at $B = 0$	$2N$ of $4N$ levels pair to $< 10^{-8}$
LO phase diagram	zero-SOC gap closes off-TRIM, 6 (μ, B) points	continuum gap $\sim 10^{-3}$–10^{-4} (grid-limited)
LO phase diagram	analytic boundary $B = \sqrt{\mu^2 + \Delta^2}$	-0.50 grid steps (finite-resolution signature)
Multi-subband builder	$M = 1$ vs. single-channel H and current I	$\max \lvert\Delta H\rvert = \max \lvert\Delta I\rvert = 0.0$
Multi-subband builder	decoupled limit vs. direct sum of M blocks	entrywise 0.0
Multi-subband builder	particle-hole symmetry, $M = 1$–4	0.0
Multi-subband builder	inter-subband selection rule (Hermitian, parity)	exact
Multi-subband builder	van Hove δ vs. diagonalized threshold spectrum	closed form 10–21% conservative
Multi-subband boundary solver	M-subband Pfaffian / B_c vs. single channel at $M = 1$	B_c difference = 0.0
Multi-subband boundary solver	uniform-current-texture reduction at $M = 2$	std/|mean| $= 1.9 \times 10^{-13}$
Finite-temperature layer	all channels at $\theta \to 0$ vs. $T = 0$	0.0 (Kubo χ: 8.9×10^{-16})
Finite-temperature layer	finite-T DC gate, $\theta = 0$–0.15	2.5×10^{-7}–4.8×10^{-6}
Disorder layer	all detectors at $W = 0$ vs. clean	bit-identical (0.0)
Disorder layer	particle-hole symmetry at $W = 5\Delta$	0.0 (any realization)
Kubo impedance	DC limit $\chi_{JJ}(0)$ vs. $d\langle J\rangle/d\phi$, 6 points	$\sim 1 \times 10^{-4}$
Parallel infrastructure	serial vs. parallel, every track	max_abs_diff = 0.0

Appendix C: Correction ledger: superseded results are preserved, not overwritten

Consistent with the data-availability statement (§II G), results that were later superseded by a more rigorous check are retained in the repository – labeled as superseded, with the correction and its reasoning recorded alongside – rather than deleted, so that the correction history is auditable rather than invisible. The load-bearing instances, each discussed at the relevant point in the main text:

1. *The threshold-detector boundary shift* ($-5.86\%/-9.36\%$) is kept alongside the threshold-free Pfaffian value ($-0.77\%/-0.60\%$ for the same self-field), because the order-of-magnitude gap between them is itself a reported result (§III A).

2. *The invariant-computation bug*, flagged in §IIB: an earlier module computing the topological invariant at the two time-reversal-invariant momenta of the Lutchyn-Oreg model built the correct real, antisymmetric Majorana-basis matrices at each momentum but never passed them to the Pfaffian routine, instead returning a hardcoded closed-form expression that happened to reproduce the correct answer only because of a parameter regime where the closed form and the genuine Pfaffian provably agree ($|4t - \mu|$ large compared to B, Δ, true everywhere this codebase's parameters are used, but not true in general). The discard was found by a line-by-line audit, traced to confirm it had never propagated into any reported boundary-shift result (all of which trace to an independent, always-genuine Pfaffian implementation), fixed to call the real Pfaffian routine, and re-validated against the analytic boundary with zero disagreements over a 1600-point sweep. We preserve this level of detail here – not because it changed any physics result in this paper, but because it is a concrete instance of the validation discipline – catching, tracing, fixing, and re-validating a bug before trusting any of its downstream numbers – that this paper's data-availability and reproducibility claims rest on.

3. *The self-consistent shift at the initially assumed coupling* ($+0.05\%$ at $g = 3 \times 10^5$) is kept, labeled, because it is what revealed the dynamical bifurcation: it is a mathematically valid but dynamically unreachable root, and preserving it makes the existence-versus-attractiveness distinction (§III B) concrete.

4. *The disorder inductive-feedback shift* passed through two recorded corrections – a search-

window coding bug, then a finite-size inflation caught by an N-independence check – with both superseded versions preserved in the repository. The main text presents the final, twice-corrected reading (§III D), and the correction process is itself part of the argument that a confident non-null claim at high disorder does not survive.

5. *The multi-subband $M = 4$ loop-gain statistic* was revised from a sparse-sampling value (3.40, one surviving point) to a properly-resolved one (2.06) after densifying that chain within the identical preregistered window; the revision is reported rather than the original quietly replaced (§III E), as a worked example of the sampling risk the preregistration was designed to guard against.

None of these corrections is presented as a footnote to be skimmed past: where the correction process is methodologically instructive, it is part of the paper's actual argument, which is that a spurious effect is dismantled only by cross-checks each more rigorous than the last.

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